Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: New Project

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 81.0 kWp

Lagolago - Philippines

Author

Project summary

Geographical Site	Situation		Project settings	
Lagolago	Latitude	10.75 °N	Albedo	0.20
Philippines	Longitude	124.80 °E		
	Altitude	38 m		
	Time zone	UTC+8		

Weather data
Lagolago
Meteonorm 8.1 (2016-2021), Sat=100% - Synthetic

System summary

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation		Near Shadings	User's needs	
Fixed plane		No Shadings	Daily profile	
Tilt/Azimuth	15 / 0 °		Constant over the year	
			Average	242 kWh/Day

System information

PV Array		Inverters	
Nb. of modules	135 units	Nb. of units	3 units
Pnom total	81.0 kWp	Pnom total	75.0 kWac
		Pnom ratio	1.080

Results summary

Produced Energy	136259 kWh/year	Specific production	1682 kWh/kWp/year	Perf. Ratio PR	83.38 %
Used Energy	88266 kWh/year			Solar Fraction SF	43.05 %

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General parameters

Grid-Connected System | **No 3D scene defined, no shadings**

PV Field Orientation

Orientation
Fixed plane
Tilt/Azimuth 15 / 0 °

Sheds configuration
No 3D scene defined

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Average Height 3.3 °

Near Shadings
No Shadings

User's needs
Daily profile
Constant over the year
Average 242 kWh/Day

Hourly load	0 h	1 h	2 h	3 h	4 h	5 h	6 h	7 h	8 h	9 h	10 h	11 h	
	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	kW
	12 h	13 h	14 h	15 h	16 h	17 h	18 h	19 h	20 h	21 h	22 h	23 h	
	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	kW

PV Array Characteristics

PV module
Manufacturer JA solar
Model JAM78-S30-600-MR
(Original PVsyst database)
Unit Nom. Power 600 Wp
Number of PV modules 135 units
Nominal (STC) 81.0 kWp

Inverter
Manufacturer Growatt New Energy
Model MID 25KTL3-X1
(Original PVsyst database)
Unit Nom. Power 25.0 kWac
Number of inverters 3 units
Total power 75.0 kWac

Array #1 - Sub-array #2
Number of PV modules 45 units
Nominal (STC) 27.00 kWp
Modules 3 string x 15 In series
At operating cond. (50°C)
Pmpp 24.66 kWp
U mpp 610 V
I mpp 40 A

Number of inverters 3 * MPPT 33% 1 unit
Total power 25.0 kWac
Operating voltage 200-1000 V
Pnom ratio (DC:AC) 1.08
No power sharing between MPPTs

Array #2 - Sub-array #2
Number of PV modules 45 units
Nominal (STC) 27.00 kWp
Modules 3 string x 15 In series
At operating cond. (50°C)
Pmpp 24.66 kWp
U mpp 610 V
I mpp 40 A

Number of inverters 3 * MPPT 33% 1 unit
Total power 25.0 kWac
Operating voltage 200-1000 V
Pnom ratio (DC:AC) 1.08
No power sharing between MPPTs

Array #3 - Sub-array #3
Number of PV modules 45 units
Nominal (STC) 27.00 kWp
Modules 3 string x 15 In series
At operating cond. (50°C)
Pmpp 24.66 kWp
U mpp 610 V
I mpp 40 A

Number of inverters 3 * MPPT 33% 1 unit
Total power 25.0 kWac
Operating voltage 200-1000 V
Pnom ratio (DC:AC) 1.08
No power sharing between MPPTs

PV Array Characteristics

Total PV power

Nominal (STC)	81 kWp
Total	135 modules
Module area	377 m²

Total inverter power

Total power	75 kWac
Number of inverters	3 units
Pnom ratio	1.08
No power sharing	

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	251 mΩ
Global wiring resistance	84 mΩ
Loss Fraction	1.5 % at STC

Module Quality Loss

Loss Fraction	-0.8 %

Module mismatch losses

Loss Fraction	2.0 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

Horizon definition

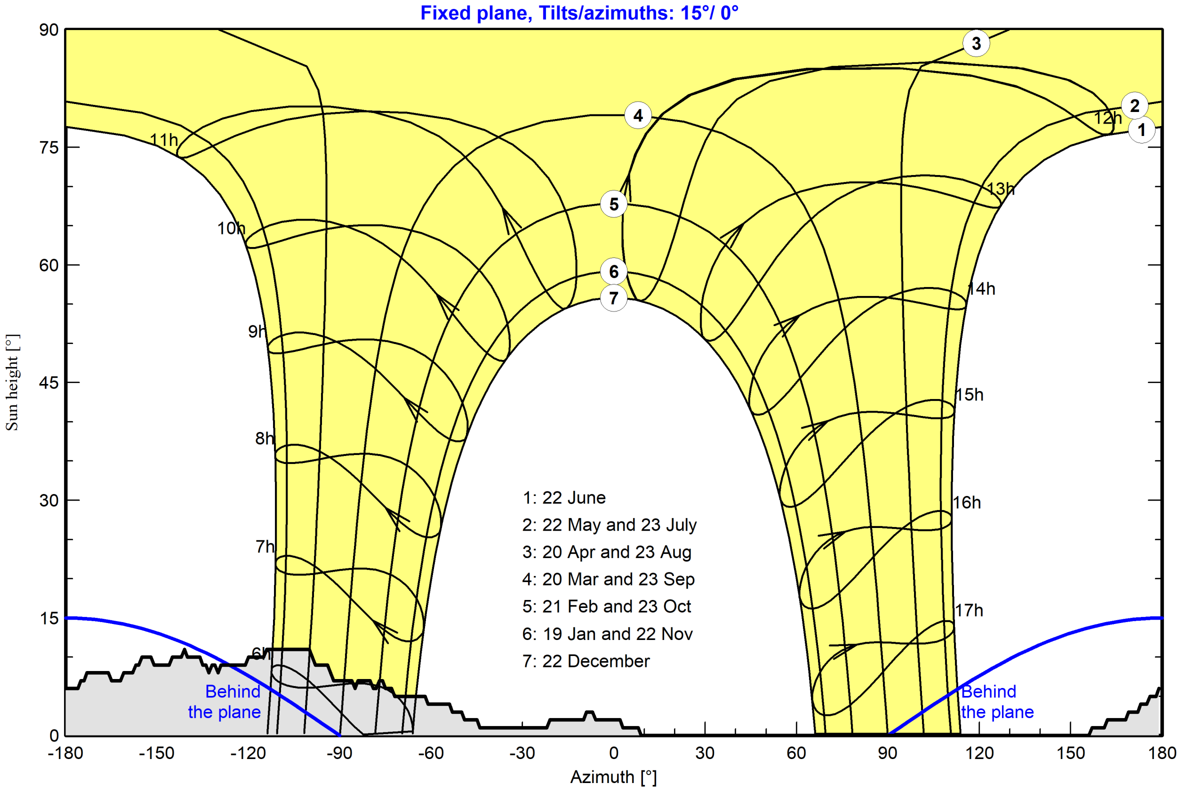
Horizon from Meteonorm web service, lat=10.746, lon=124.7963

Average Height	3.3 °	Albedo Factor	0.94
Diffuse Factor	0.99	Albedo Fraction	100 %

Horizon profile

Azimuth [°]	-180	-176	-174	-173	-162	-161	-158	-156	-155	-152	-146	-145	-142	-140
Height [°]	6.0	6.0	7.0	8.0	7.0	8.0	8.0	9.0	10.0	10.0	9.0	10.0	10.0	10.0
Azimuth [°]	-136	-135	-133	-132	-131	-129	-121	-120	-114	-100	-98	-92	-86	-85
Height [°]	10.0	9.0	8.0	9.0	9.0	9.0	9.0	10.0	11.0	11.0	9.0	7.0	7.0	6.0
Azimuth [°]	-80	-79	-77	-76	-75	-72	-62	-61	-53	-50	-49	-44	-22	-21
Height [°]	7.0	6.0	7.0	7.0	6.0	5.0	5.0	4.0	3.0	3.0	2.0	1.0	1.0	2.0
Azimuth [°]	-9	-7	-6	3	8	9	157	161	162	172	173	176	178	179
Height [°]	3.0	3.0	2.0	1.0	1.0	0.0	1.0	1.0	2.0	3.0	4.0	5.0	5.0	6.0

Sun Paths (Height / Azimuth diagram)

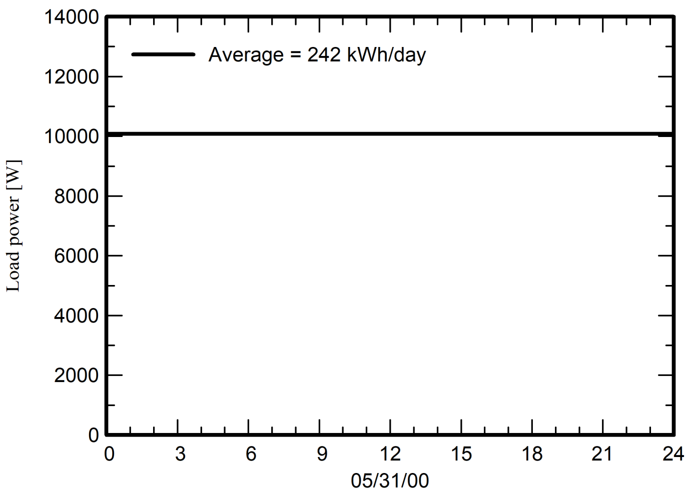


Detailed User's needs

Daily profile, Constant over the year, average = 242 kWh/day

Hourly load	0 h	1 h	2 h	3 h	4 h	5 h	6 h	7 h	8 h	9 h	10 h	11 h	
	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	kW
	12 h	13 h	14 h	15 h	16 h	17 h	18 h	19 h	20 h	21 h	22 h	23 h	
	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	10.08	kW

Daily profile



Main results

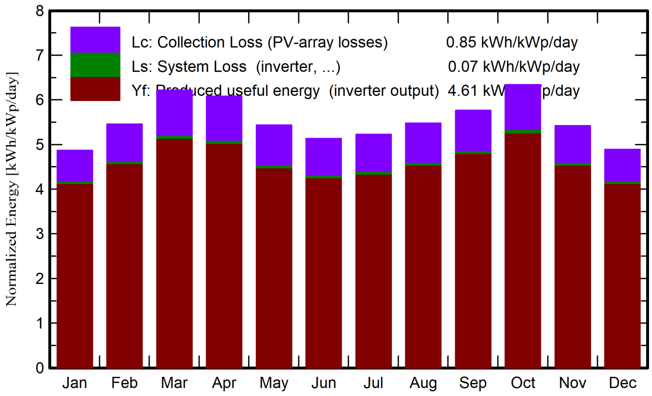
System Production

Produced Energy (P50)	136259 kWh/year	Specific production (P50)	1682 kWh/kWp/year	Perf. Ratio PR	83.38 %
Produced Energy (P90)	121248 kWh/year	Specific production (P90)	1497 kWh/kWp/year	Solar Fraction SF	43.05 %
Produced Energy (P95)	117025 kWh/year	Specific production (P95)	1445 kWh/kWp/year		

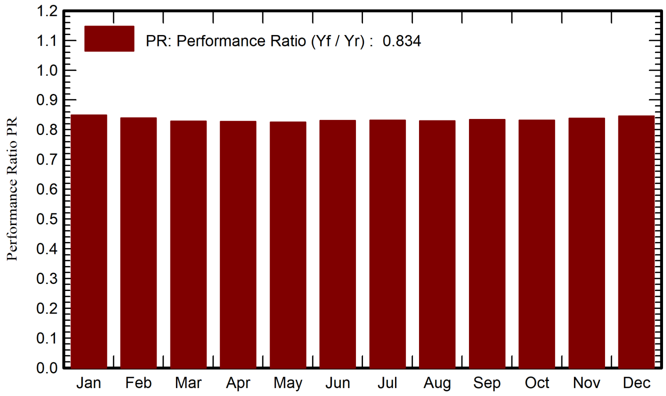
Economic evaluation

Investment		Yearly cost		LCOE	
Global	1,475,365.00 PHP	Annuities	0.00 PHP/yr	Energy cost	0.83 PHP/kWh
Specific	18.2 PHP/Wp	Run. costs	56,795.31 PHP/yr		
		Payback period	1.5 years		

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

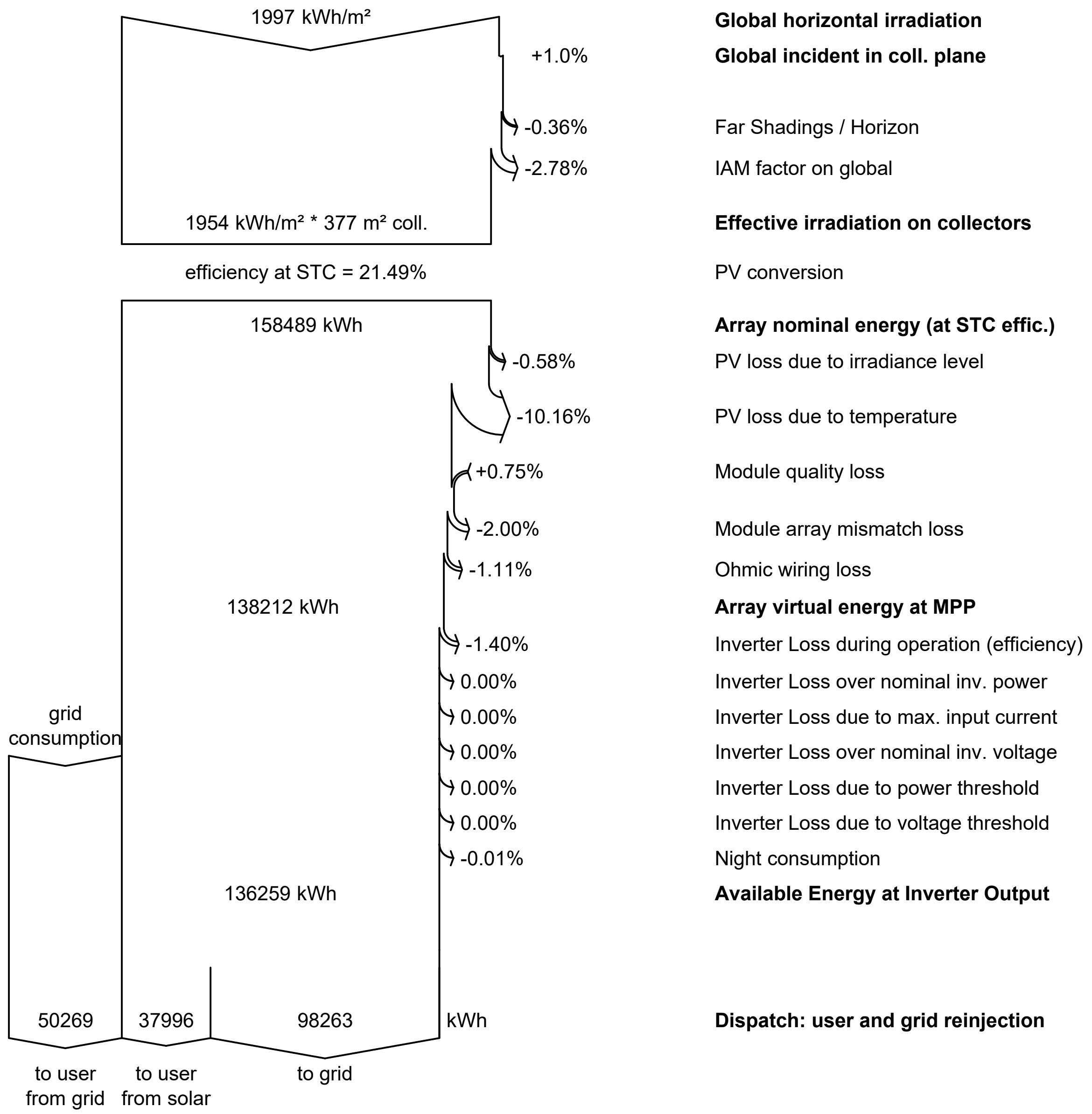
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January	136.7	73.02	25.52	151.1	147.0	10538	7497	3050	7337	4447
February	142.3	62.97	26.12	152.8	149.0	10540	6771	2765	7620	4006
March	187.3	73.90	27.27	192.7	187.2	13119	7497	3248	9685	4249
April	187.8	77.07	28.27	182.6	176.7	12409	7255	3203	9032	4051
May	181.3	75.43	29.32	168.6	162.7	11431	7497	3292	7971	4204
June	169.2	74.71	28.49	154.0	148.2	10518	7255	3183	7180	4071
July	176.8	81.91	28.42	162.1	155.7	11067	7497	3253	7659	4243
August	178.1	81.01	28.50	169.8	164.0	11578	7497	3282	8134	4214
September	172.6	79.54	27.47	173.0	167.6	11854	7255	3222	8467	4033
October	184.4	70.95	27.38	196.6	191.4	13426	7497	3326	9918	4170
November	146.0	62.35	26.58	162.8	158.0	11203	7255	3082	7966	4173
December	134.6	70.01	26.14	151.6	146.9	10529	7497	3089	7293	4408
Year	1997.1	882.87	27.46	2017.5	1954.3	138212	88266	37996	98263	50269

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	E_Grid	Energy injected into grid
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid

PVsyst V7.4.7
VC0, Simulation date:
06/06/26 21:10
with V7.4.7

Loss diagram



PVsyst V7.4.7
VC0, Simulation date:
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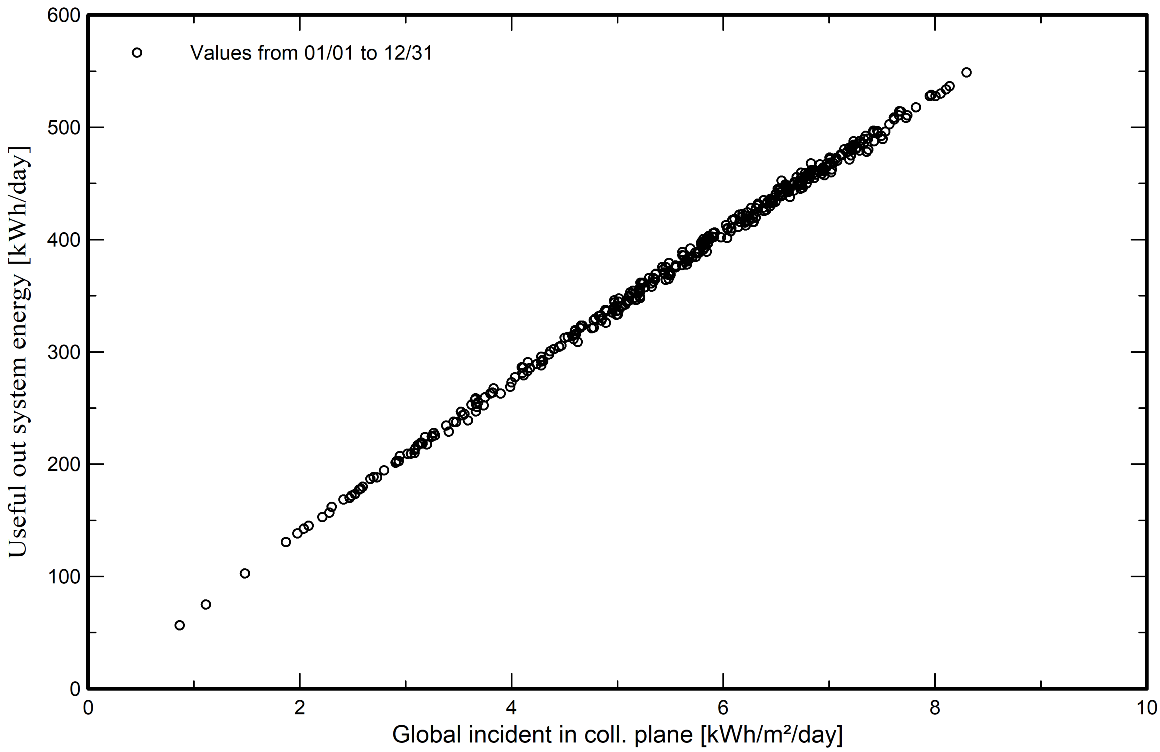
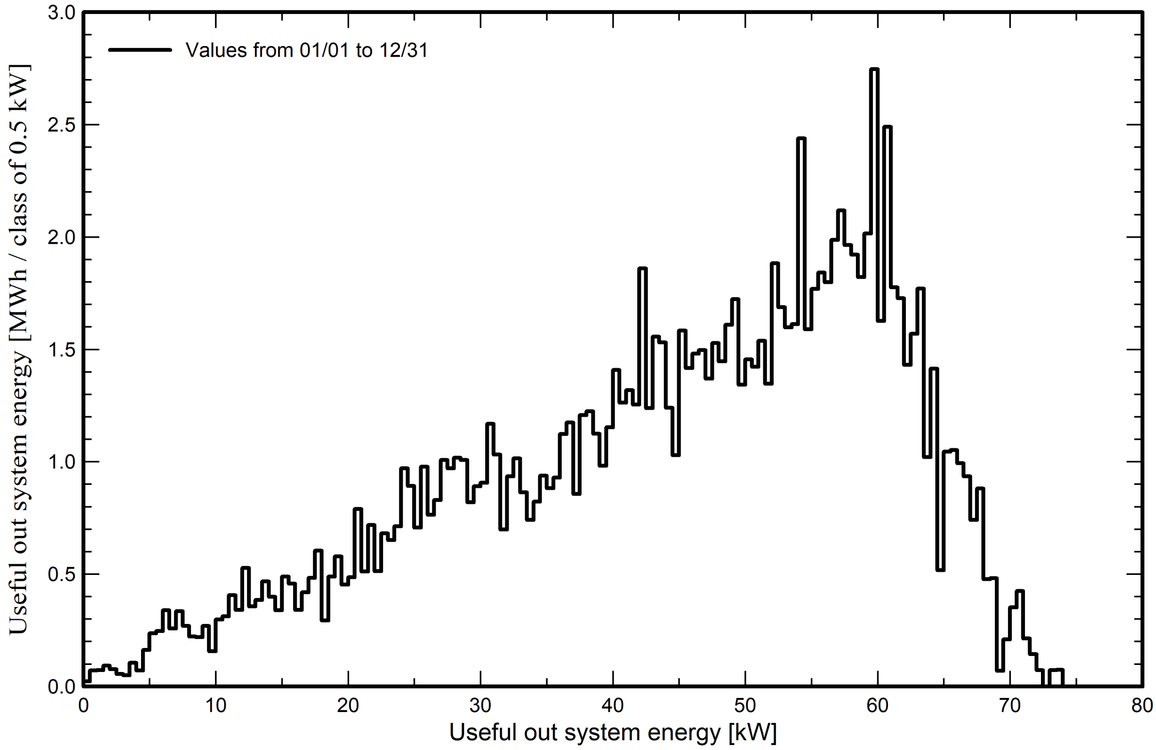
with V7.4.7

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Meteonorm 8.1 (2016-2021), Sat=100%
Kind	Monthly averages
Synthetic - Multi-year average	
Year-to-year variability(Variance)	8.4 %

Specified Deviation

Climate change	0.0 %

Global variability (weather data + system)

Variability (Quadratic sum)	8.6 %

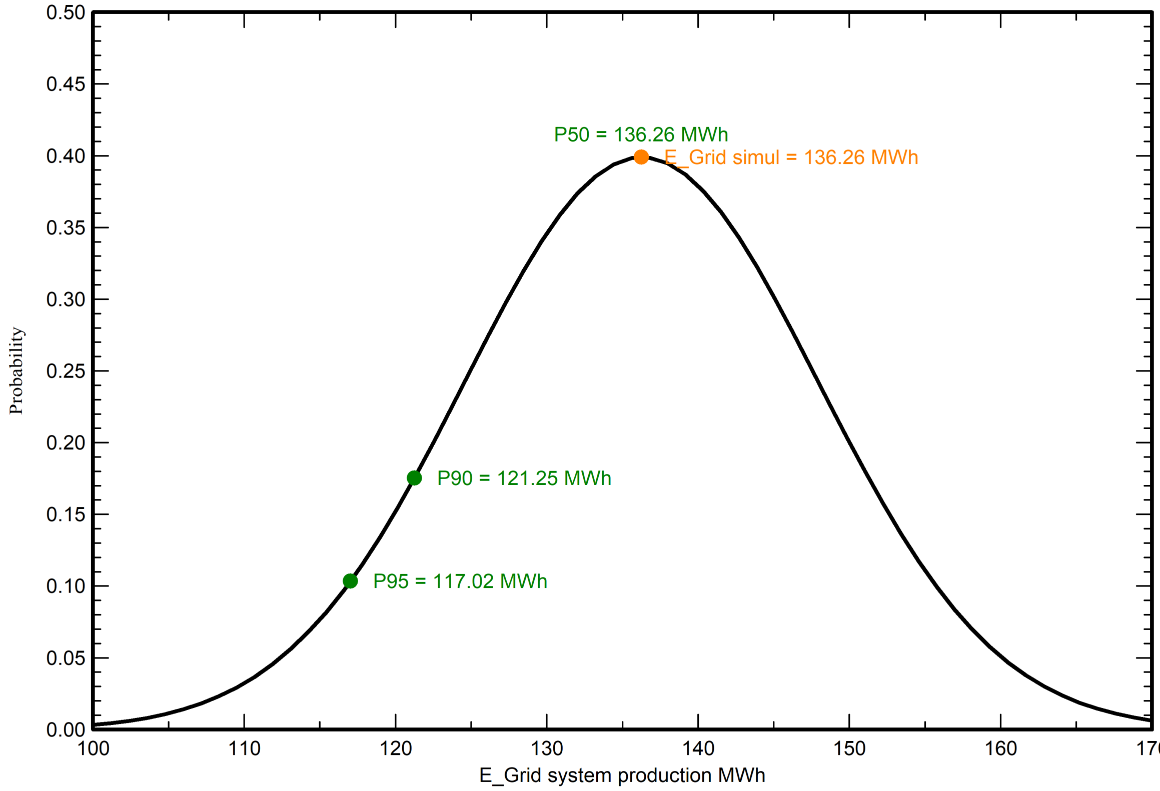
Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

Annual production probability

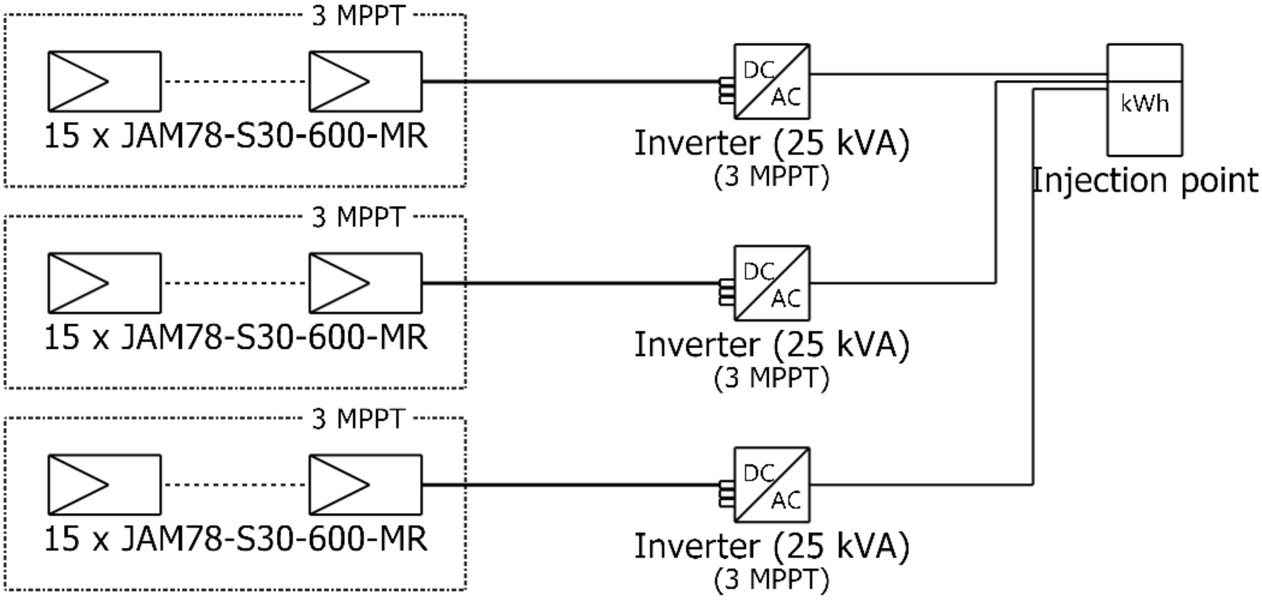
Variability	11.71 MWh
P50	136.26 MWh
P90	121.25 MWh
P95	117.02 MWh

Probability distribution



Single-line diagram

PVsyst V7.4.7
VC0, Simulation date:
06/06/26 21:10
with V7.4.7



PV module	JAM78-S30-600-MR
Inverter	MID 25KTL3-X1
String	15 x JAM78-S30-600-MR

	New Project	
	VC0 : New simulation variant	06/06/26

Cost of the system

Installation costs

Item	Quantity units	Cost PHP	Total PHP
PV modules			
JAM78-S30-600-MR	135	8,319.00	1,123,065.00
Supports for modules	72	900.00	64,800.00
Inverters			
MID 25KTL3-X1	3	70,000.00	210,000.00
Other components			
Accessories, fasteners	1	20,000.00	20,000.00
Wiring	1	5,000.00	5,000.00
Combiner box	1	30,000.00	30,000.00
Monitoring system, display screen	1	9,000.00	9,000.00
Surge arrester	1	10,000.00	10,000.00
Studies and analysis			
Engineering	1	2,000.00	2,000.00
Permitting and other admin. Fees	1	1,500.00	1,500.00
		Total	1,475,365.00
		Depreciable asset	1,417,865.00

Operating costs

Item	Total PHP/year
Maintenance	
Provision for inverter replacement	42,000.00
Total (OPEX)	42,000.00
Including inflation (2.00%)	56,795.31

System summary

Total installation cost	1,475,365.00 PHP
Operating costs (incl. inflation 2.00%/year)	56,795.31 PHP/year
Useful energy from solar	38.0 MWh/year
Energy sold to the grid	98.3 MWh/year
Cost of produced energy (LCOE)	0.8303 PHP/kWh

Financial analysis

Simulation period

Project lifetime	30 years	Start year	2027

Income variation over time

Inflation	2.00 %/year
Production variation (aging)	0.00 %/year
Discount rate	1.00 %/year

Income dependent expenses

Income tax rate	0.00 %/year
Other income tax	0.00 %/year
Dividends	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (PHP)	Depreciable (PHP)
PV modules				
JAM78-S30-600-MR	Straight-line	20	0.00	1,123,065.00
Supports for modules	Straight-line	20	0.00	64,800.00
Inverters				
MID 25KTL3-X1	Straight-line	30	0.00	210,000.00
Accessories, fasteners	Straight-line	20	0.00	20,000.00
		Total	0.00	1,417,865.00

Financing

Own funds	1,475,365.00 PHP

Electricity sale

Feed-in tariff	5.50000 PHP/kWh
Duration of tariff warranty	20 years
Annual connection tax	0.00 PHP/kWh
Annual tariff variation	0.0 %/year
Feed-in tariff decrease after warranty	0.00 %

Self-consumption

Consumption tariff	12.50000 PHP/kWh
Tariff evolution	0.0 %/year

Return on investment

Payback period	1.5 years
Net present value (NPV)	23,287,359.46 PHP
Internal rate of return (IRR)	65.90 %
Return on investment (ROI)	1578.4 %

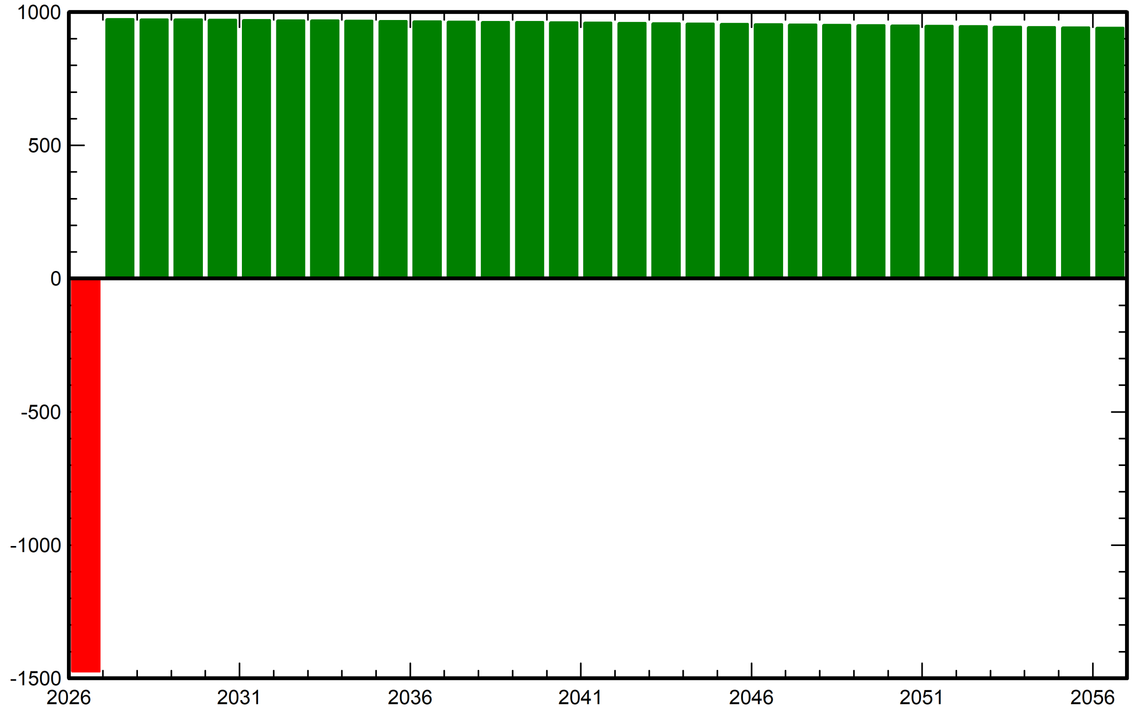
Financial analysis

Detailed economic results (PHP)

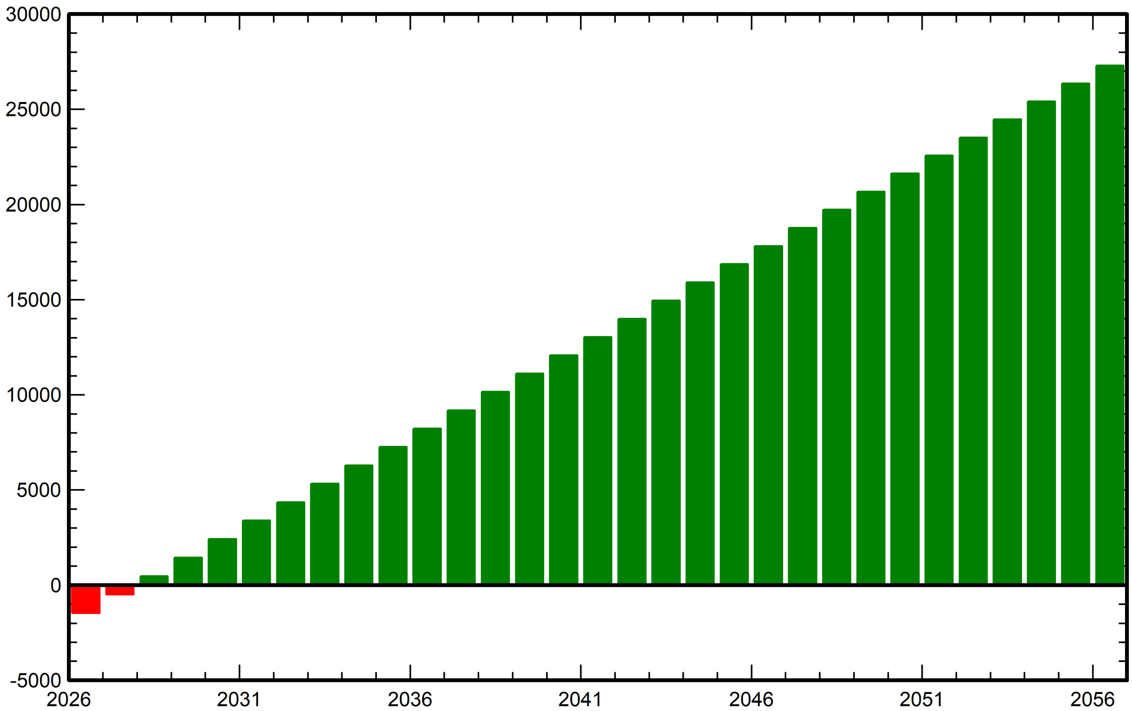
Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Self-cons. saving	Cumul. profit	% amorti.
0	0	1,475,365	0	0	0	0	0	0	-1,475,365	0.0%
1	540,518	0	42,000	67,393	431,125	0	498,518	474,956	-511,529	65.3%
2	540,518	0	42,840	67,393	430,285	0	497,678	474,956	441,941	130.0%
3	540,518	0	43,697	67,393	429,428	0	496,822	474,956	1,385,138	193.9%
4	540,518	0	44,571	67,393	428,554	0	495,948	474,956	2,318,158	257.1%
5	540,518	0	45,462	67,393	427,663	0	495,056	474,956	3,241,091	319.7%
6	540,518	0	46,371	67,393	426,754	0	494,147	474,956	4,154,030	381.6%
7	540,518	0	47,299	67,393	425,826	0	493,220	474,956	5,057,064	442.8%
8	540,518	0	48,245	67,393	424,880	0	492,274	474,956	5,950,285	503.3%
9	540,518	0	49,210	67,393	423,916	0	491,309	474,956	6,833,779	563.2%
10	540,518	0	50,194	67,393	422,931	0	490,325	474,956	7,707,635	622.4%
11	540,518	0	51,198	67,393	421,927	0	489,321	474,956	8,571,939	681.0%
12	540,518	0	52,222	67,393	420,903	0	488,297	474,956	9,426,776	738.9%
13	540,518	0	53,266	67,393	419,859	0	487,252	474,956	10,272,233	796.3%
14	540,518	0	54,331	67,393	418,794	0	486,187	474,956	11,108,391	852.9%
15	540,518	0	55,418	67,393	417,707	0	485,100	474,956	11,935,335	909.0%
16	540,518	0	56,526	67,393	416,599	0	483,992	474,956	12,753,147	964.4%
17	540,518	0	57,657	67,393	415,468	0	482,861	474,956	13,561,906	1019.2%
18	540,518	0	58,810	67,393	414,315	0	481,708	474,956	14,361,694	1073.4%
19	540,518	0	59,986	67,393	413,139	0	480,532	474,956	15,152,589	1127.0%
20	540,518	0	61,186	67,393	411,939	0	479,332	474,956	15,934,671	1180.0%
21	540,518	0	62,410	7,000	471,109	0	478,109	474,956	16,708,016	1232.5%
22	540,518	0	63,658	7,000	469,860	0	476,860	474,956	17,472,702	1284.3%
23	540,518	0	64,931	7,000	468,587	0	475,587	474,956	18,228,804	1335.5%
24	540,518	0	66,230	7,000	467,289	0	474,289	474,956	18,976,397	1386.2%
25	540,518	0	67,554	7,000	465,964	0	472,964	474,956	19,715,555	1436.3%
26	540,518	0	68,905	7,000	464,613	0	471,613	474,956	20,446,351	1485.9%
27	540,518	0	70,284	7,000	463,235	0	470,235	474,956	21,168,859	1534.8%
28	540,518	0	71,689	7,000	461,829	0	468,829	474,956	21,883,149	1583.2%
29	540,518	0	73,123	7,000	460,395	0	467,395	474,956	22,589,292	1631.1%
30	540,518	0	74,585	7,000	458,933	0	465,933	474,956	23,287,359	1678.4%
Total	**16,215,554**	**1,475,365**	**1,703,859**	**1,417,865**	**13,093,829**	**0**	**14,511,694**	**14,248,676**	**23,287,359**	**1678.4%**

Financial analysis

Yearly net profit (kPHP)



Cumulative cashflow (kPHP)



CO_2 Emission Balance

Total: 1626.0 tCO_2

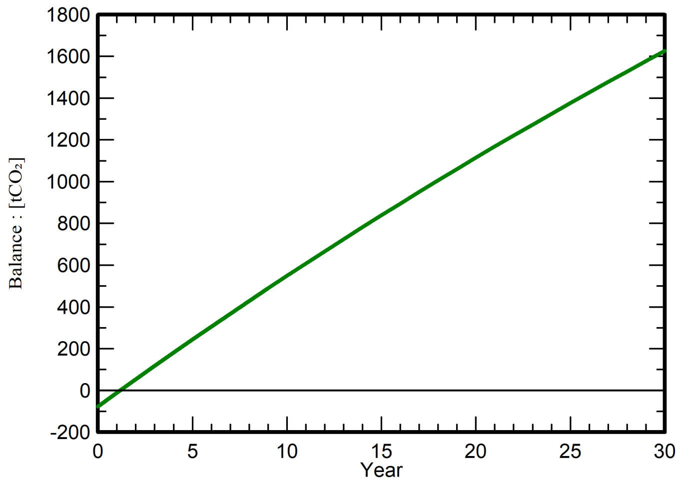
Generated emissions

Total: 76.61 tCO_2
Source: Detailed calculation from table below

Replaced Emissions

Total: 1962.3 tCO_2
System production: 136.27 MWh/yr
Grid Lifecycle Emissions: 480 gCO_2/kWh
Source: IEA List
Country: Philippines
Lifetime: 30 years
Annual degradation: 1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[kgCO_2]
Modules	1713 kgCO2/kWp	43.2 kWp	73990
Supports	3.20 kgCO2/kg	720 kg	2305
Inverters	317 kgCO2/units	1.00 units	317